

Exit IRR / MOIC

Sensitivity Analyzer

- A professional LBO return modeling tool built for the current private equity market.
- I built this tool to account for floating-rate debt, buyer-constrained exit pricing, and different market condition scenario analysis.

WHY THIS TOOL EXISTS

The Rate Cycle Problem

PE deals originated in 2019–2021 at 4–5% debt costs now face 9–10% refinancing rates. Standard LBO models use a fixed rate — masking the true return impact.

Buyer Constraints

Rising rates don't just increase your debt cost; they reduce what any buyer can afford to pay. Exit multiples compress because the next buyer's LBO math no longer works at the old price.

The Liquidity Crunch

GPs are holding assets longer to avoid selling at distressed prices, creating pressure on LP distributions. This tool makes the tradeoffs quantitative and defensible.

KEY FEATURES

Floating Rate Debt

Hedged or unhedged toggle. Unhedged rates ramp linearly from entry to current — matching real SOFR drift. Hedge cost subtracted from FCF annually.

Buyer's LBO Engine

Solves for the maximum price a rational buyer can pay to hit their target IRR (default 20%), given current rates and tightened leverage limits.

Bull / Base / Bear Scenarios

Bull = competitive auction at theoretical max. Base = -12% discount. Bear = -25% — stressed market, few bidders. Shows the realistic return range.

Rate Erosion Quantified

Every output compares floating rate vs. original rate environment, isolating the pure cost of the rate cycle on your equity returns.

OUTPUTS AT A GLANCE

Sensitivity Grid

Color-coded IRR and MOIC across every combination of exit multiple (8–16x) and hold period (3–7 years). Toggle between rate environments.

Visual Charts

IRR line chart and MOIC bar chart with market cap line — shows exactly which exit multiples are above what the market will realistically support.

Buyer's LBO Table

For each hold year: implied exit multiple, your IRR, and your MOIC in each scenario. Toggle between Bull / Base / Bear with one click.

BULL CASE

Competitive auction. Buyer pays theoretical maximum. Full leverage available.

BASE CASE

Normal process friction. -12% discount to theoretical max. Realistic clearing price.

BEAR CASE

Stressed market. Few bidders. Lender conservatism. -25% discount to max.

RATE EROSION

Every output shows original rate vs. floating rate IRR delta — pure rate damage isolated.

BUILT WITH

Next.js 14 · TypeScript · Tailwind CSS
Recharts · Newton-Raphson IRR Solver

LIVE APP

pe-analytics.vercel.app